Combining Swin Transformer With UNet for Remote Sensing Image Semantic Segmentation

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***Abstract*— Remote sensing semantic segmentation plays a significant role in various applications such as environmental monitoring, land use planning, and disaster response. Convolutional neural networks (CNNs) have been dominating remote sensing semantic segmentation. However, due to the limitations of convolution operations, CNNs cannot effectively model global context. The success of transformers in the natural language processing (NLP) domain provides a new solution for global context modeling. Inspired by the Swin transformer, we propose a novel remote sensing semantic segmentation model called CSTUNet. This model employs a dual-encoder structure consisting of a CNN-based main encoder and a Swin transformer-based auxiliary encoder. We first utilize a detail-structure preservation module (DPM) to mitigate the loss of detail and structure information caused by Swin transformer downsampling. Then we introduce a spatial feature enhancement module (SFE) to collect contextual information from different spatial dimensions. Finally, we construct a position-aware attention fusion module (PAFM) to fuse contextual and local information. Our proposed model obtained 70.75% mean intersection over union (MIoU) on the ISPRS-Vaihingen dataset and 77.27% MIoU on the ISPRS-Potsdam dataset.**

***Index Terms*— Feature fusion, remote sensing image, semantic segmentation, Swin transformer.**

I. Introduction

DUE to the rapid development of remote sensing and satellite technologies, acquiring remote sensing image data has become increasingly easy and abundant. These remote sensing data hold significant value in various fields [1], [2] such as urban planning [3], [4], disaster assessment [5], agricultural production [6], military evaluation, and environmental monitoring [7]. Consequently, developing efficient and reliable methods to extract useful information from remote sensing images is of utmost importance.

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Remote sensing semantic segmentation [8] is a widely applied computer vision technique in the field of remote sensing image processing. Its goal is to assign pixels in remote sensing images to their corresponding semantic categories (e.g., buildings, forests, and roads). This technique not only helps in interpreting and understanding remote sensing images but also promotes the development of various remote sensing applications.

In recent years, because of the rapid development of deep learning technologies, numerous models based on convolutional neural networks (CNNs) have emerged in the field of image semantic segmentation. Fully convolutional networks (FCNs) [9] significantly improved the accuracy of semantic segmentation by replacing fully connected layers with convolutional layers and introducing skip connections. Subsequently, models based on encoder–decoder architectures were developed, taking full advantage of the superior performance of the encoder–decoder structure. Among them, the U-Net [10] model preserved more details and semantic information by passing low-level features to high-level layers through skip connections, leading to the creation of numerous U-Net-based variant models [11], [12], [13]. By combining encoder–decoder structure and atrous spatial pyramid pooling (ASPP) modules, the DeepLabV3+ [14] network achieved accurate handling of object boundaries and detail information.

Despite the notable accomplishments of deep CNN-based models on various datasets, their limited receptive field due to small convolutional kernels gives them an advantage in extracting local information while constraining their ability to model global context. Remote sensing images often have complex scenes where objects within the same category may have large feature differences and objects from different categories may have similar features. Researchers have proposed attention mechanisms to address this issue. SENet [15] emphasized important feature channels through channel-wise weighting, thus improving model performance. CBAM [16] focused on key features through spatial and channel attention to enhance the model's ability to process remote-sensing images. LANet [51] enhances low-dimensional and high-dimensional features by patch attention. DANet [17] achieved more effective global feature information capture through two parallel attention modules. Moreover, to capture feature information at different scales, researchers have employed multiscale feature fusion strategies. For example, PSPNet [18] and UperNet [19] used pyramid modules to obtain feature information at different scales, thereby improving the performance of remote sensing image semantic segmentation.

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In recent years, the transformer architecture, which has achieved remarkable results in the natural language processing (NLP) field [20], has also begun to attract attention and research in the vision domain. Vision transformer [21] successfully applied the transformer architecture to the visual field, achieving considerable success in image classification, a visual task. Building on this foundation, the Swin transformer [22] and DETR [23] have achieved success in image segmentation and object detection tasks, respectively.

In this article, we propose a hybrid remote sensing image semantic segmentation model that combines CNN and transformer, aiming to address the limitations of CNN in terms of global context modeling capabilities. In the model design, we adopt an encoder–decoder architecture and construct an auxiliary encoder based on the Swin transformer auxiliary encoder, as well as a main encoder based on CNN. Meanwhile, by employing skip connections, we achieve feature fusion and information exchange between the encoders and the decoder. Specifically, we use the detail-structure preservation module (DPM) attached after Swin transformer downsampling and design a position-aware attention fusion module (PAFM) to aggregate global context and local information. We attach the spatial feature enhancement module (SFE) between the Swin transformer to enhance its context modeling capability.

The contributions of this article are as follows.

1) To address the issue of small-scale feature information loss during downsampling in the Swin transformer, we design the DPM: this module introduces multiscale information and dilated convolution, which helps the model extract richer feature representations while preserving the details of the original image.
2) We introduce the spatial feature enhancement module (SFE) in the auxiliary encoder, which collects context information of different spatial dimensions through pooling convolution operations in horizontal and vertical directions. Combined with a Swin transformer block, the SFE helps compensate for the shortcomings of a Swin transformer in context modeling, thereby improving the performance of remote sensing semantic segmentation.
3) To more effectively integrate the global features extracted from the auxiliary encoder and the local features extracted from the main encoder, we design a PAFM. This module introduces position-sensitive channel attention mechanisms, enabling precise localization and recognition of the target of interest. This fusion strategy helps to enhance the accuracy and robustness of remote sensing image segmentation.

II. Related Work

A. CNN-Based Semantic Segmentation of Remote Sensing

Since Audebert et al. [24] applied FCN to remote sensing image processing, CNN-based methods have dominated the field of remote sensing semantic segmentation [25], [26], [27], [28]. However, the decoder structure of FCN is not sensitive enough to details when recovering image resolution, leading to less precise segmentation results. To address this issue, UNet introduced an encoder–decoder architecture. UNet's contracting path reduces the resolution and extracts features through the basic structure of CNNs, while the expanding path restores the resolution through upsampling and obtains contextual information via skip connections. Thanks to the efficiency of the encoder–decoder structure, it has gradually become a common architecture for remote sensing semantic segmentation. However, due to the limitations of convolution kernels, convolutional networks are not adept at global context modeling.

To compensate for this shortcoming, Zhao et al. [18] developed the pyramid scene parsing network, which combines multiscale contextual information to improve performance. Li et al. [60] propose a lightweight CNN with a spatial path and a contextual path. FarSeg++ [54] enhances the connection between foregrounds and enhances the difference between foreground and background through the foreground-scene relation module. Li et al. [55] proposed a gating mechanism that selectively fused multiple levels of features. DGMN [56] achieves significant computational complexity reduction through dynamic graph passing networks. SFNet [57] learns the semantic flow between adjacent scale feature maps through the flow alignment module and is integrated with the FPN structure. Cheng et al. [14] adopted DeepLabV3 [29] as the encoder, utilizing encoder–decoder and ASPP to enhance network performance. Researchers have also introduced attention mechanisms. Hu et al.'s [15] SE attention module adjusted the weights of each channel by learning the relationships between feature channels. CBAM [16] focused on key features by combining spatial attention and channel attention. Ding et al. [51] propose a patch attention module to enhance the embedding of contextual information, and enrich the semantic information of the underlying features by embedding the local focus of the high-level features through the attention embedding module. DANet [17] incorporated position attention modules and channel attention modules to capture global context information. Mou et al. [53] proposed S-RA-FCN, which learns and deduces the global relationship between any two spatial locations or feature maps through the spatial relationship module and the channel relationship module and then generates a relationship-enhanced feature representation.

B. Transformer-Based Semantic Segmentation of Remote Sensing Images

The transformer uses self-attention mechanisms to track relationships between sequences, and learning context and surpass previous recurrent and CNN models in the field of machine translation. Researchers have been attracted to the transformer because of its ability to encode long-term dependencies and learn efficient feature representations. Dosovitskiy et al. [21] achieved excellent results in image classification tasks by dividing images into a series of 2-D image patches and treating them as sequences for processing. In this context, researchers have begun to explore the application of transformers to semantic segmentation tasks [30], [31], [32], [33]. SETR [34] approached the semantic segmentation task as a sequence-to-sequence problem and built a model by integrating a transformer encoder with a straightforward decoder. However, ViT itself has a large

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number of parameters, and its columnar structure is not suitable for dense prediction tasks. Building on this, PVT [35] introduced CNN's pyramid structure to reduce the token count at different stages. Segformer [36] adopted a hierarchical decoder structure, outputting multiscale features. Swin transformer [22] utilized sliding windows and hierarchical structures window strategy. By substituting the conventional multihead self-attention (MSA) with window-based MSA (W-MSA) and shifted window MSA (SW-MSA), it confines the computational complexity to a window, thus attaining linear computational complexity. This improvement effectively reduces the computation requirements of transformers, making them more feasible for remote sensing image semantic segmentation tasks.

Cao et al. [37] constructed the first pure transformer-based UNet medical image segmentation network using the Swin transformer to build an encoder-decoder architecture. DS-TransUNet [38] integrated Swin transformer features at different scales, demonstrating its effectiveness on four medical image datasets [39], [40], [41], [42].

C. Hybrid Models of Transformer and CNN

Despite the widespread attention given to the powerful global modeling capabilities of transformers, we cannot ignore the advantages of CNNs in extracting local information. As a result, Transfuse [43] and TransUNet [44] proposed hybrid models that combine transformers and CNNs, finding that relying solely on transformer networks does not yield ideal results. These transformer-based networks treat images as a series of patches, thereby neglecting the structural information between images. To fully utilize the advantages of both methods, transfuse adopted a parallel structure and integrated the strengths of the two approaches through a multimodal feature fusion strategy. On the other hand, TransUNet employed a stacked structure, gradually upsampling to restore the original resolution. STransFuse [48] uses the Swin transformer and ResNet to extract features and fuse them, respectively. He et al. [50] hierarchically integrated the Swin transformer's global dependencies into the features of CNNs through a relational aggregation module. Zhang et al. [59] used the Swin transformer to extract features and employed a ASPP block based on depth-wise separable convolution [separable atrous spatial pyramid pooling (SASPP)] in the middle to obtain a multiscale context. BANet [61] proposed a bilateral perception network that used ResT as the dependency path and utilized linear attention to aggregate the texture path of the dependency path and the stacked convolution operation.

Building upon this, we propose an auxiliary encoder based on the Swin transformer, which provides global contextual information for the main encoder with ResNet50 [45] as its backbone network. By doing so, we construct a high-performing network model for remote sensing image semantic segmentation tasks.

III. Method

In this part, we present the comprehensive architecture of the suggested CSTUNet model and subsequently provide an in-depth explanation of each individual component.

A. Network Structure

The overall architecture of CSTUNet is shown in Fig. 1. CSTUNet follows the excellent skip-connection encoder–decoder structure of UNet and adopts a hybrid model of the Swin transformer and CNN in the encoder part. CSTUNet constructs a dual-encoder based on ResNet and Swin transformer, fusing the features of the two encoder branches through the PAFM. We also designed a detail preservation module and a spatial feature enhancement module (SFE) to improve the performance of the Swin transformer.

For a given image $x \in \mathbb{R}^{H\times W\times C}$, it is fed into the main encoder and the Swin transformer-based auxiliary encoder, where H denotes the image height, W stands for the image width, and C signifies the number of image channels. In the Swin transformer-based encoder, we set the patch size to 8×8. These patches are projected to C_1 dimensions and then fed into the Swin transformer blocks.

We place a DPM before each patch token enters the Swin transformer block, which consists of W-MSA and SW-MSA. We pass the vectors through the SFE before they enter the Swin transformer block and add a residual connection to the vectors that have gone through the Swin transformer block. The SFE module collects contextual information from different spatial dimensions by performing pooling convolution operations in both horizontal and vertical directions, compensating for the limitations of window-oriented self-attention computation. We define the outputs of the four feature extraction stages of the auxiliary encoder as $S_n, n = 1, 2, 3, 4$. The output dimensions are $S_n \in \mathbb{R}^{H/2^{n+1}\times W/2^{n+1}\times(2^{n-1}\times C_1)}$, where $C_1 = 96$. Meanwhile, the image size is reduced by half and fed into the ResNet50 network. We define the outputs of the four feature extraction stages of the main encoder as $Z_n, n = 1, 2, 3, 4$. The output dimensions are $Z_n \in \mathbb{R}^{H/2^{n+1}\times W/2^{n+1}\times(2^{n-1}\times C_2)}$, where $C_2 = 128$. We use the PAFM to fuse the features from the main encoder and the auxiliary encoder together and feed them into the main encoder. In the encoding stage, we obtain the final output $F \in \mathbb{R}^{H/32\times W/32\times 1024}$. In the encoding stage, we obtain the final output, and then we use a transposed convolution to increase the resolution, adopting skip connections to connect the features from the decoder and the encoder. The prediction results are obtained by continuous upsampling.

B. Swin Transformer

The conventional transformer module is primarily composed of a few essential elements, such as MSA, multilayer perceptron (MLP), and layer normalization (LN). With the MSA mechanism, a token is required to calculate self-attention for every other token, resulting in a quadratic increase in the model's computational complexity. For computationally intensive tasks like semantic segmentation, this cost is too high. Swin transformer introduced W-MSA and SW-MSA, which divide the image into nonoverlapping windows and perform self-attention computation within the windows, with each window containing (MxM) patches. As shown in Fig. 2,

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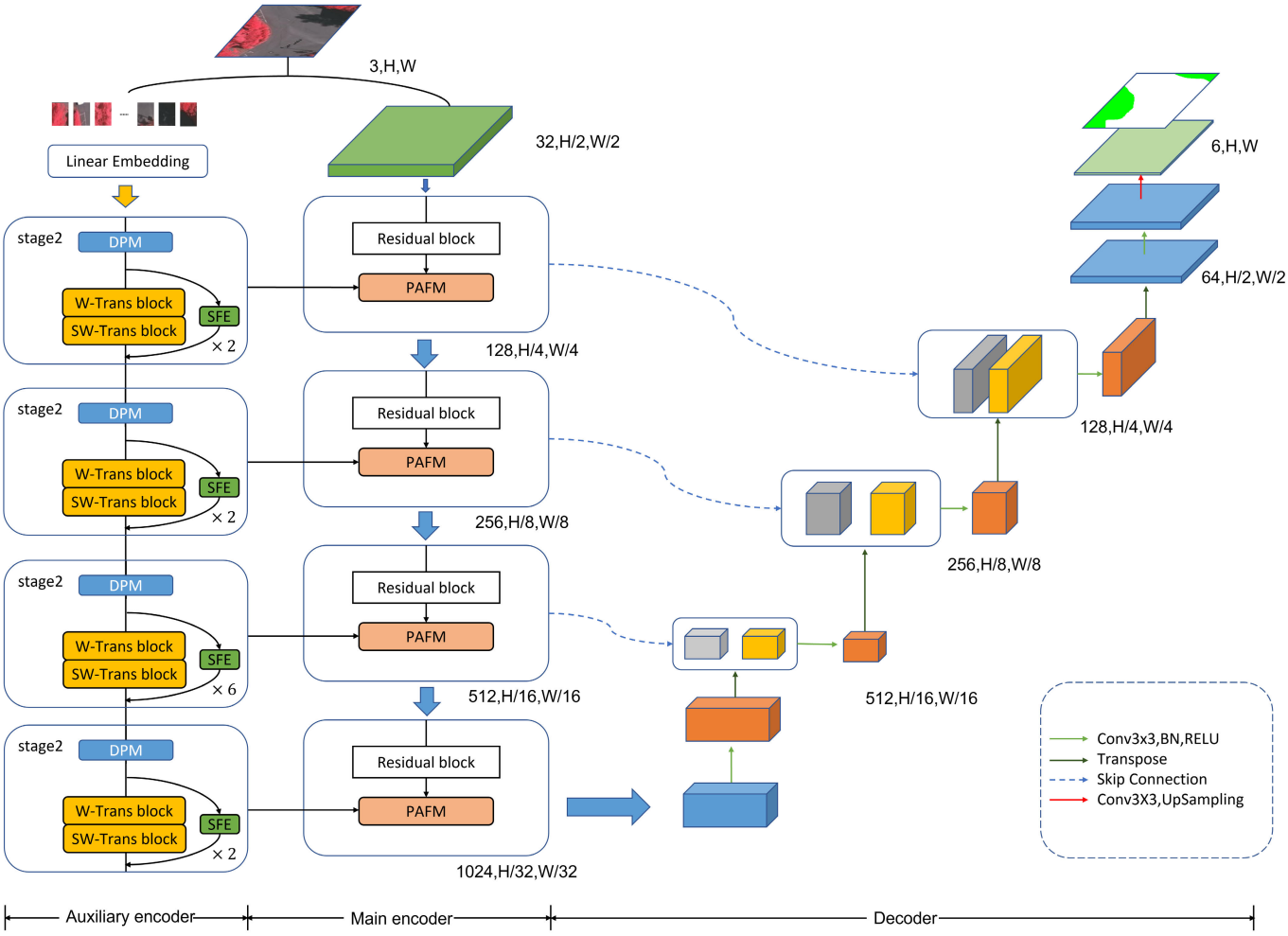


Fig. 1. Overall structure of CSTUNet.

the computation is as follows:

$$\begin{aligned}\hat{x}^{l} &= \text{W-MSA}\left(\text{LN}\left(x^{l-1}\right)\right) + x^{l-1}\\ x^{l} &= \text{MLP}\left(\text{LN}\left(\hat{x}^{l}\right)\right) + \hat{x}^{l}\\ \hat{x}^{l+1} &= \text{SW-MSA}\left(\text{LN}\left(x^{l}\right)\right) + x^{l}\\ x^{l+1} &= \text{MLP}\left(\text{LN}\left(\hat{x}^{l+1}\right)\right) + \hat{x}^{l+1}\end{aligned} \tag{1}$$

where $\hat{x}^{l}$ and $\hat{x}^{l+1}$ represent the output of W-MSA and SW-MSA, x^{l} and x^{l+1} denote the output after passing through the MLP layer.

C. Detail-Structure Preservation Module

In the Swin transformer, the model reduced the computation cost by downsampling patches through projection or patch merging operations. However, remote sensing images themselves have a rich amount of information and complex natural scenes, which may result in the loss of a significant amount of detail and structural information due to these downsampling operations. To alleviate this issue and improve segmentation accuracy, we introduce the DPM after the downsampling process to retain more detailed information and extract richer feature representations.

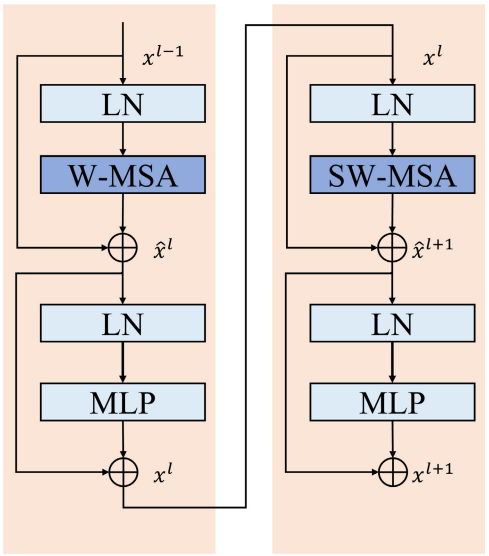


Fig. 2. Swin transformer block.

As shown in Fig. 3, we first reshape the vector into a feature map form, then feed the feature map into a

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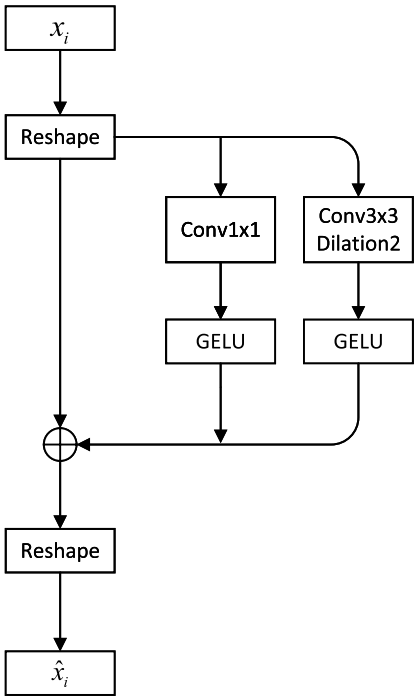


Fig. 3. Structure of DPM.

3×3 dilated convolution (dilation factor $= 2$) and a 1×1 convolution layer, respectively. The feature maps are added together through a GELU activation function, combined with the original feature map using residual connections, and finally reshaped back into a vector form. The mathematical representation is as follows:

$$\text{out} = \text{GELU}(\text{DConv}(x)) \oplus \text{GELU}(\text{Conv}(x)) \oplus x \tag{2}$$

where $\oplus$ stands for element-level addition.

D. Spatial Feature Enhancement Module

Although the Swin transformer effectively reduces memory consumption compared to the vision transformer by performing self-attention computation within windows, its global modeling capability is still somewhat limited, even when using a window-shift strategy. To address this issue, we introduce an improved strip pooling module (SPM) [46]. The SPM collects long-range contextual information in two spatial dimensions through strip convolutions. Since matrix multiplication can be computationally intensive, we optimize the SPM by adding ReLU activation functions and residual connections to reduce computational complexity and mitigate the vanishing gradient issue. We refer to this optimized module as the spatial feature enhancement module (SFE). The structure of the SFE is shown in Fig. 4.

Reshaping the given $s^{l-1} \in \mathbb{R}^{h \times w \times c_n}$ into $g_{i,j} \in \mathbb{R}^{c_n \times h \times w}$. Perform pooling operations on $g_{i,j}$ separately along the vertical and horizontal dimensions, followed by convolution operations, and finally add the features from both dimensions together. This can be represented mathematically as follows:

$$y_i^h = \frac{1}{h} \sum_{0<=i<h} g_{i,j}, \quad y_j^w = \frac{1}{w} \sum_{0<=j<w} g_{i,j} \tag{3}$$

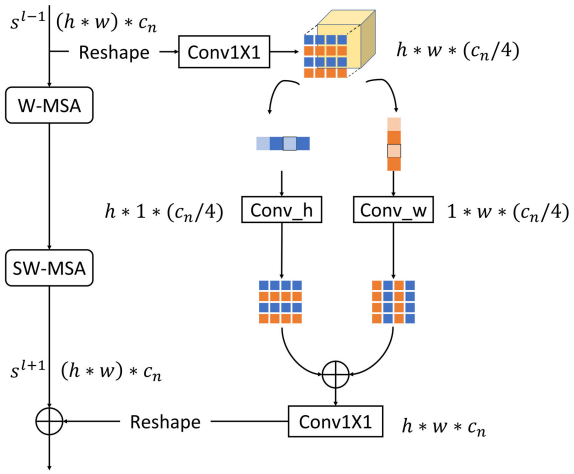


Fig. 4. Structure of SFE.

$$y = \text{Conv3} \times 3\left(f\left(\text{Conv}h\left(y_i^h\right)\right) \oplus f\left(\text{Conv}w\left(y_j^w\right)\right)\right) \tag{4}$$

$$F = s^{l+1} \oplus \varphi(y) \tag{5}$$

where $c_n = 2^{n-1}C_1, h = (H/2^{n+1}), w = (W/2^{n+1})$, i, and j are indicators in the vertical and horizontal directions, respectively, Conv$h$ represents a convolution kernel (3, 1) with a step size of 1 and zero filling of (1, 0), Conv$w$ corresponds to it accordingly, f denotes the upsampling operation, and φ represents the 1×1 convolution operation with ReLU activation function.

E. Position-Aware Attention Fusion Module

To effectively fuse features from the main encoder and the Swin transformer auxiliary encoder, we design PAFM. As shown in Fig. 5, this module adopts a position-aware channel attention mechanism, which can precisely locate and recognize the targets of interest, thereby enhancing the global and local information captured by both encoders. Finally, the obtained features are fused with the features extracted by the encoders. Specifically, we implement feature fusion through the following formula:

$$\hat{z}_i = \text{CA}(Z_i), \quad \hat{s}_i = \text{CA}(S_i) \tag{6}$$

$$f_i = \text{Conv3} \times 3(\text{Concat}([\hat{z}_i \otimes Z_i, \hat{s}_i \otimes S_i, Z_i \oplus S_i])) \tag{7}$$

where Z_i denotes the output from the feature extraction stage of the main encoder, S_i represents the output of the auxiliary encoder feature extraction stage, and CA is coordinate attention [47]. Conv3 $\times$ 3 represents a convolution operation with kernel size $= 3 \times 3$, stride $= 1$, and padding $= 1$. The final generated features f_i effectively combine both global and local information.

IV. Experiments

A. Datasets

1) Vaihingen: In the Vaihingen dataset, there are 33 images with an average size of 2496 $\times$ 2064 pixels and a GSD of 9 cm. We test using ground truth (GT) where object boundaries are not eroded by three black pixels. We use 23 images as the training set (image IDs: 1, 2, 3, 4, 6, 7, 12, 13, 14, 16,

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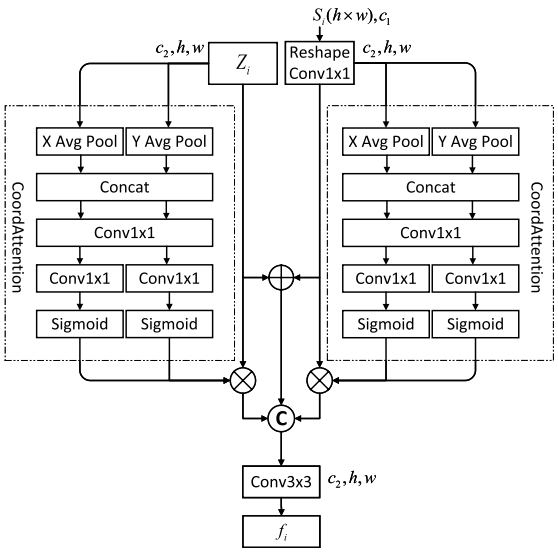


Fig. 5. Structure of PAFM.

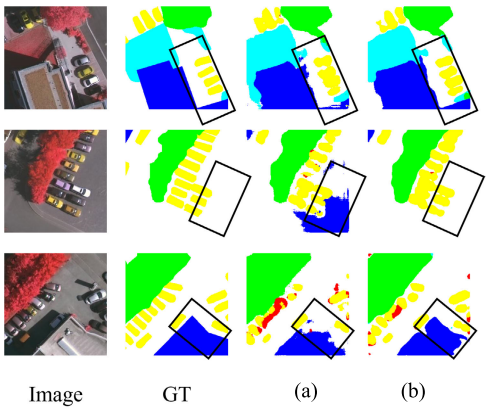


Fig. 6. Comparison of the baseline before and after using the DPM. (a) baseline and (b) baseline + DPM.

17, 21, 22, 23, 26, 27, 29, 31, 32, 33, 35, and 37), and the remaining ten images as the test set (image IDs: 5, 8, 10, 11, 15, 20, 24, 28, 30, 34, and 38). We split the images into 256 × 256 patches and employ data augmentation techniques for the training set, such as random horizontal flipping and random rotation.

2) Potsdam: In the Potsdam dataset, there are 38 images with a size of 6000 × 6000 pixels and a GSD of 5 cm. We use RGB images as the dataset. Following previous works [48], [49], [50], we use 14 images as the test set (image IDs: 2_13, 2_14, 3_13, 3_14, 4_13, 4_14, 4_15, 5_13, 5_14, 5_15, 6_13, 6_14, 6_15, and 7_13) and 24 images as the training set (image IDs: 2_10, 2_11, 2_12, 3_10, 3_11, 3_12, 4_10, 4_11, 4_12, 5_10, 5_11, 5_12, 6_7, 6_8, 6_9, 6_10, 6_11, 6_12, 7_7, 7_8, 7_9, 7_10, and 7_11). We also use the GT without boundaries for testing and adopt the same data augmentation techniques as in the Vaihingen dataset.

Due to the very small proportion of the clutter/background class in both datasets, we ignore this category when performing quantitative evaluations after [51] and [52].

B. Training Settings

We use SGD as the optimizer, setting the momentum to 0.9, weight decay to 1e−4, initial learning rate to 0.1, and adopting the "poly" learning rate decay strategy. All experiments are conducted on a computer equipped with an NVIDIA RTX 3090 GPU. During the training process, we set the batch size to 16 and the maximum number of iterations (epochs) to 100. To address the class imbalance issue in the Vaihingen and Potsdam datasets, we employ a combination of cross-entropy loss L_C and Dice loss L_D to supervise the model training. The composite loss function L can be described as follows:

$$L = 0.5L_C + 0.5L_D. \tag{8}$$

C. Evaluation Metrics

We assess our model's effectiveness using mean intersection over union (MIoU) and mean F1 scores. These evaluation measures rely on the confusion matrix, containing true positives (TPs), false positives (FPs), true negatives (TNs), and false negatives (FNs). The IoU calculation formula for each category is as follows:

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}. \tag{9}$$

The formula for calculating the F1 score for each category is defined as follows:

$$F1 = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \tag{10}$$

where recall = (TP/(TP + FN)), precision = (TP/(TP+FP)), MIoU is the average of all classes of IoU, and mean F1 is the average of all classes of F1.

D. Ablation Experiments

To assess the effectiveness of our suggested network architecture, we employ a CNN with Swin-transformer assistance as the baseline model and conduct ablation experiments on the Vaihingen dataset. In our network, the auxiliary encoder adopts Swin-T with an initial projection of $C_1 = 96$, a window size of 8, layer numbers of {2, 2, 6, 2}, and corresponding multihead numbers of {3, 6, 12, 24}.

1) Influence of Detail-Structure Preservation Module: As can be seen from Table I, MIoU increased by 1.43% and mean F1 increased by 1.01%. Compared to the baseline, the model with DPM shows the most significant improvement in the "Car" class, with an IoU increase of 4.79%. As shown in Fig. 6, in the first two lines, the segmentation results of the model with DPM are much clearer than those of the baseline. In the third row, the car next to the tree is identified as clutter/background in the baseline, while it is much better recognized in the model with DPM compared to the baseline. This result indicates that adopting DPM preserves more detail and structural information.

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TABLE I
ABLATION STUDY RESULTS FOR THE PROPOSED MODULE ON THE VAIHINGEN DATASET

Method	Module name			IoU(%)					Evaluation indicators	
	DPM	SFE	PAFM	Imp. Surf.	Building	Low Veg.	Tree	Car	MIoU(%)	Mean F1(%)
swin-baseline				70.63	78.74	55.34	71.69	54.38	66.16	79.22
baseline				70.87	78.69	59.56	72.64	54.01	67.15	79.99
baseline+DPM	√			71.81	79.68	60.16	73.46	58.80	68.58	81.00
baseline+SFE		√		73.82	81.30	59.14	73.14	56.32	68.74	81.09
baseline+PAFM			√	73.96	81.29	58.21	73.52	56.25	68.64	81.46
baseline+DPM+SFE	√	√		74.18	81.35	60.31	74.16	59.20	69.83	81.93
baseline+DPM+PAFM		√	√	74.98	82.07	60.25	73.46	59.08	69.96	82.00
baseline+SFE+PAFM	√		√	74.27	81.94	59.78	73.72	58.60	69.66	81.78
baseline+DPM+SFE+PAFM	√	√	√	75.16	83.00	61.45	73.27	60.88	70.75	82.58

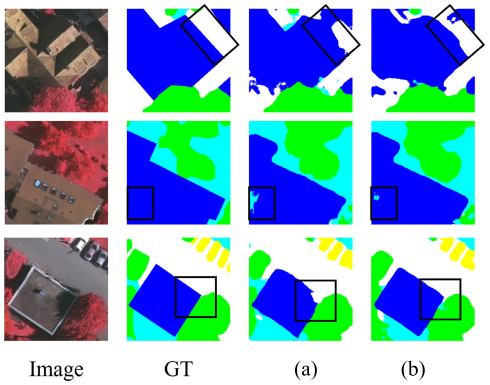


Fig. 7. Comparison of the baseline before and after using the SFE module. (a) baseline and (b) baseline + SFE.

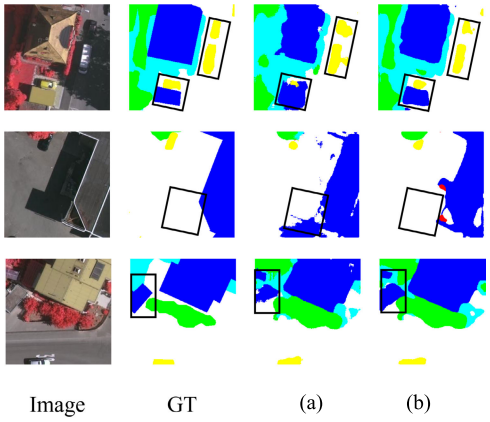


Fig. 8. Comparison of the baseline before and after using the PAFM. (a) baseline and (b) baseline + PAFM.

2) Influence of Spatial Feature Enhancement Module: As shown in Table I, after using SFE, the MIoU and mean F1 scores increased by 1.59% and 1.10%, respectively, which validates the effectiveness of SFE in the model. The opaque water surface increased by 2.95%, and buildings increased by 2.61%. In the first row of Fig. 7, the opaque water surface is sandwiched between two buildings. After using SFE, it effectively avoids confusion in recognizing the two categories. In the third row, it identifies the parts of the buildings that are obscured by trees. In practice, using the SFE to collect spatial context information reduces the impact of closely connected objects.

3) Influence of PAFM: As seen in Table I, after using PAFM, the MIoU and mean F1 scores increased by 1.49% and 1.47%, respectively. Except for the "low vegetation", all other categories showed improvement. From the visualization results in Fig. 8, the segmentation results for each class are significantly improved. In the first and third rows, compared to the baseline, the segmentation performance of cars and buildings obscured by shadows is significantly enhanced. In the second row, the segmentation accuracy of the opaque water edge is improved. This fully demonstrates the effectiveness of the PAFM.

4) Result on Different Seeds: Upon detailed analysis in Table II, it is evident that, despite utilizing different random seeds, their impact on the experimental results is relatively minimal (in the article, the random seed value of 1234 was used). This suggests that our method demonstrates stability and is not significantly influenced by minor variations in the initial random seed.

E. Comparison With Other Methods

We conducted experiments on two datasets, Vaihingen and Potsdam, and compared CSTUNet with several existing methods, including FCN, UNet, DeepLabV3+, DANet, PSPNet, and TransUNet. All of these methods use an untrained ResNet50 as their backbone.

1) Results on Vaihingen Dataset: In the Vaihingen dataset, Table III lists the numerical results for each model in detail. From the results, our CSTU achieves higher MIoU and mean F1 on the Vaihingen dataset than other models. LANet uses the block attention mechanism to obtain the best performance in the CNN network. DeepLabV3+ uses ASPP to obtain multiscale information, achieving the second-best performance in the CNN model. Meanwhile, the network model of STUNet specifically used for remote sensing images adopts a hybrid model of CNN and Swin transformer, and MIOU is higher than other models.

Compared to the STUNet model, CSTUNet's MIoU and mean F1 have increased by 1.61% and 1.2%, respectively. As seen in Fig. 9, in the second row, compared to STUNet, between building and low vegetation, CSTUNet distinguishes two categories with similar characteristics, tree, and low vegetation, although the classification of opaque water surface is not as effective as the CNN model, and it is also better than the STUNet hybrid model. CSTUNet distinguishes between the two similar classes of trees and shrubs. Although the segmentation performance of the opaque water surface class is not as good as CNN models, it is still better than the mixed TransUNet model. In the third

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TABLE II
COMPARISON RESULTS USING DIFFERENT RANDOM SEEDS IN THE VAIHINGEN DATASET

Seed	IoU(%)					Evaluation indicators	
	Imp. Surf.	Building	Low Veg.	Tree	Car	MIoU(%)	MeanF1(%)
56	75.26	83.51	61.48	73.21	59.47	70.59	82.43
1234(paper)	75.16	83.00	61.45	73.27	60.88	70.75	82.58

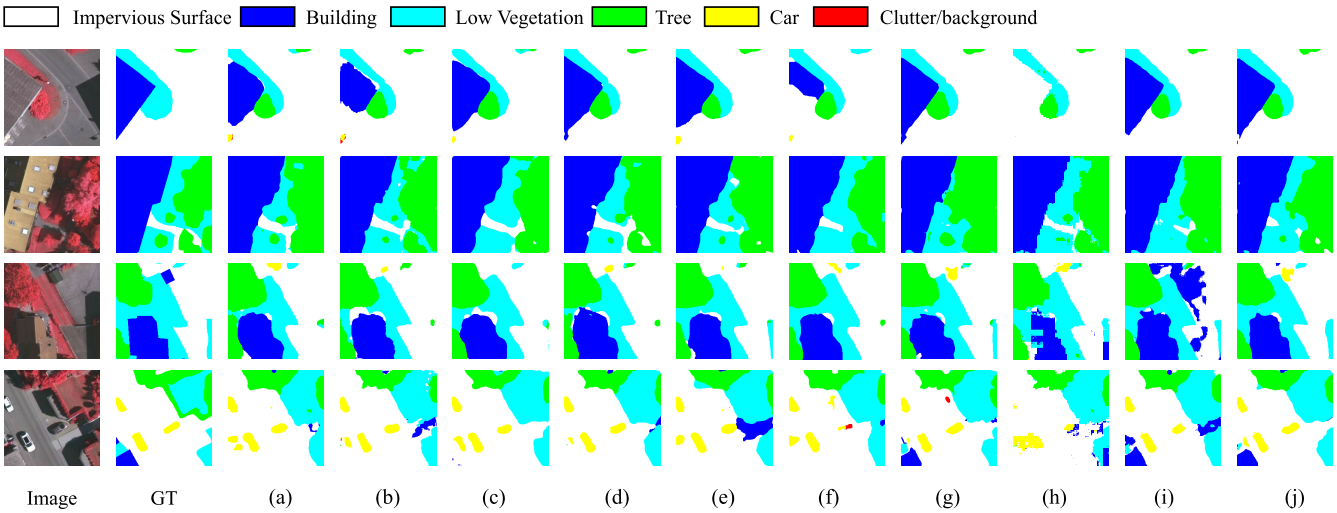


Fig. 9. Visualization of segmentation on the Vaihingen dataset. a: FCN, b: UNet, c: DANet, d: DeepLabV3+, e: PSPNet, f: LANet, g: TransUNet, h: SwinUNet, i: STUNet, and j: CSTUNet.

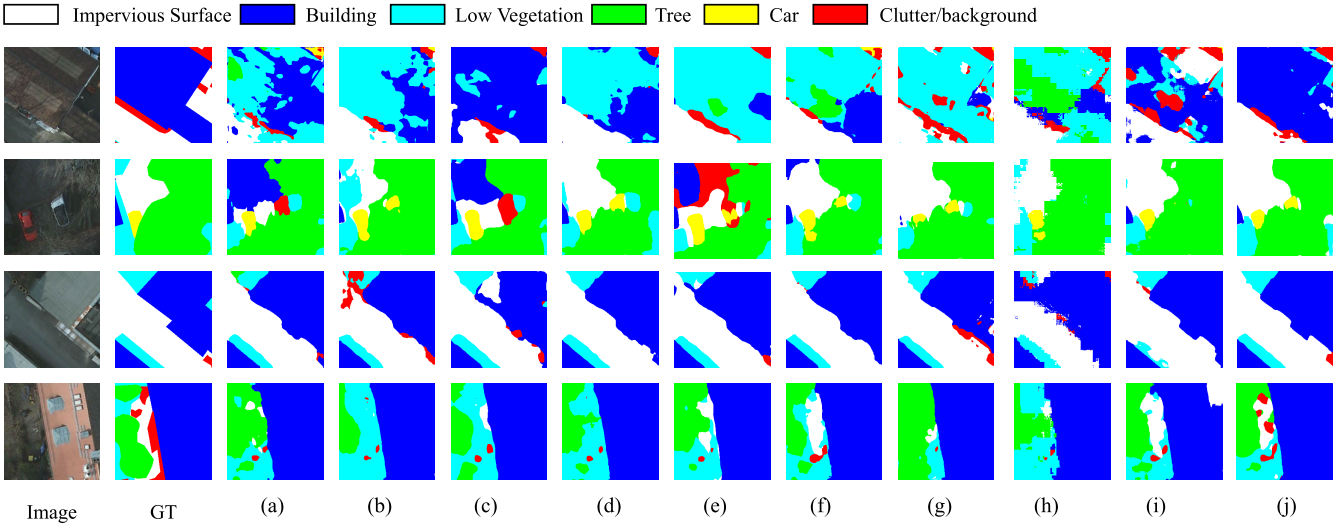


Fig. 10. Visualization of segmentation on the Potsdam dataset. (a) FCN, (b) UNet, (c) DANet, (d) DeepLabV3+, (e) PSPNet, (f) LANet, (g) TransUNet, (h) SwinUNet, (i) STUNet, and (j) CSTUNet.

row, CSTUNet correctly identifies the opaque water surface between trees and low vegetation. Although some models are also recognized, such as DeepLabV3+ and TransUNet, our model segmentation is better than theirs. This indicates that CSTUNet reduces the recognition error between highly similar objects and effectively combines global contextual information and local information, resulting in noticeable segmentation improvements for occluded objects.

2) Results on Potsdam Dataset: As shown in Table IV, the CSTUNet proposed in this article achieves an MIoU of 77.27% and a mean F1 of 86.95% on the Potsdam dataset, significantly outperforming other methods. Our approach achieves an IoU score of 88.91% in the Building class, which is more than 2.4% higher than other networks. From the visualization results in Fig. 10, in the first row, due to the shadow of the shrubs cast on the building, all models except DeepLabV3+ identify a large area of the building as shrubs. In contrast, our model and DeepLabV3+ correctly recognize it as a building, with our model demonstrating better segmentation performance than DeepLabV3+. Thanks to our model's superior integration of global contextual features and local features, it accurately identifies the opaque water surface sandwiched between shrubs and buildings in the third row.

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TABLE III
COMPARISON OF SEGMENTATION OUTCOMES USING THE VAIHINGEN DATASET

Method	IoU(%)					Evaluation indicators	
	Imp. Surf.	Building	Low Veg.	Tree	Car	MIoU(%)	MeanF1(%)
FCN [9]	72.03	78.77	59.06	72.67	45.76	65.65	78.61
UNet [10]	71.29	78.69	59.90	71.98	49.81	66.33	79.28
DANet [17]	72.82	79.44	60.14	72.76	48.44	66.72	79.48
DeepLabV3+ [14]	72.55	79.72	59.53	72.72	49.79	66.86	79.62
PSPNet [18]	72.42	79.49	59.66	71.92	45.82	65.86	78.76
LANet [51]	74.87	80.40	57.35	72.97	50.24	67.17	79.78
TransUNet [44]	73.95	82.07	58.84	73.21	54.49	68.51	80.86
SwinUNet [44]	64.51	68.23	52.94	69.88	29.49	57.01	71.32
STUNet [50]	74.40	81.61	57.18	73.70	58.82	69.14	81.38
CSTUNet	75.16	83.00	61.45	73.27	60.88	70.75	82.58

TABLE IV
COMPARISON OF SEGMENTATION OUTCOMES USING THE POTSDAM DATASET

Method	IoU(%)					Evaluation indicators	
	Imp. Surf.	Building	Low Veg.	Tree	Car	MIoU(%)	MeanF1(%)
FCN [9]	74.79	80.37	65.15	63.17	77.23	72.14	83.63
UNet [10]	77.00	84.03	65.77	63.68	78.04	73.72	84.62
DANet [17]	77.89	83.57	67.83	65.40	77.52	74.44	85.17
DeepLabV3+ [14]	79.34	85.18	69.16	65.51	78.84	75.60	85.91
PSPNet [18]	77.52	83.10	66.91	64.49	75.73	73.55	84.57
LANet [51]	78.21	83.23	65.70	64.84	77.56	73.91	84.79
TransUNet [44]	78.60	85.56	67.27	66.76	76.56	74.95	85.49
SwinUNet [44]	71.18	77.54	60.06	55.46	70.02	66.85	79.85
STUNet [50]	78.82	86.49	68.48	65.97	77.83	75.52	85.84
CSTUNet	80.66	88.91	70.52	66.62	79.68	77.27	86.95

TABLE V
COMPARISON OF MODEL PARAMETERS AND SPEED

Method	Parameters	Speed (FPS)	
		Vaihingen	Potsdam
FCN	22.70M	368	370
UNet	25.13M	207	211
DANet	46.72M	96	101
DeepLabV3+	38.48M	59	60
PSPNet	24.32M	197	205
LANet	23.81M	147	169
transUNet	105.43M	23	26
SwinUNet	41.39M	45	52
STUNet	160.97M	11	12
CSTUNet	117.77M	18	20

3) Speed Analysis: Table V lists the parameter count and running speed of the models under the same operating environment. The model's speed indicates the quantity of images processed within a given time frame, quantified in frames per second (FPS). Models with transformer structures do not have an advantage in terms of parameter count and running speed compared to CNN structures. Our CSTUNet is not the slowest on both datasets. Compared to STUNet, our network parameters are less than it and the results are better than it. Despite these drawbacks, we believe that CSTUNet still has research value in remote sensing semantic segmentation.

V. CONCLUSION

In this article, we address the inadequacy of CNNs in modeling global context information by constructing a hybrid semantic segmentation model with a CNN-based main encoder and a Swin-transformer-based branch encoder. Our proposed DPM module utilizes dilated convolutions to retain more detailed features during the downsampling process. The SFE module establishes vertical and horizontal feature extraction, eliminating the loss of contextual spatial information caused by the window mechanism. The PAFM module fuses contextual and local information. However, our model exhibits noticeable jagged edges in the boundary regions during recognition, which are not smooth and have a large parameter count, limiting its applicability. In future work, we will focus on compressing the model and addressing the issue of unsmooth edges.

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